

Global spectral truncation for the direct marked identity

Complete analytic argument; independently reviewed

30 September 2026

Status. Full independent mathematical reviews by the root and size-bounds agents passed. A separate independent-directions review also passed, including the appended unmarked and off-annulus results. This is not formal verification.

The important truncation is in *differential order*, before the infinite spectral tail is estimated. No estimate for the complete high-order polarization of the infinite Poisson kernel is asserted.

Use the notation of `independent_directions/marked_cancellation_identity.tex`: $m = M + 1$, $\ell = \lfloor M/4 \rfloor$, $N = M - \ell$, $z_j = \ell(1 - p_j)$, and $s = \sum_{j \neq i} (1 - p_j)$. The row weights w form a probability measure. A uniformly chosen ℓ -subset B of the unmarked columns has weight $A_B = \prod_{j \in B} p_j$, complement R , and mean $v = N^{-1} \sum_{j \in R} z_j$. Put

$$a_k = \frac{e_k((z_j - v)_{j \in R})}{\binom{N}{k}}, \quad a_0 = 1, \quad a_1 = 0.$$

All expectations below include the uniformly chosen subset before multiplication by A_B ; thus no division by its possibly small mean is present. We use the established finite-balance and row-mass bounds in the following exact form, with constants independent of m :

$$\max_{j \in [m]} z_j \leq C_0(s + 1), \quad \mathbb{E}_w e^{-bs} \leq C_b/m \quad (b > 0 \text{ fixed}). \quad (1)$$

These follow from the reviewed endpoint-balance and row-mass lemmas, including after deleting one column. For the marked column itself, the full-family estimate is rearranged to absorb its own contribution to the deficit sum, for all sufficiently large m .

For a smooth function F define the finite differential functional

$$\mathcal{D}_J F = (z_i - v)F(v) - \sum_{k=2}^J \left[\frac{(v - z_i)F^{(k)}(v)}{k!} + \frac{F^{(k-1)}(v)}{(k-1)!} \right] a_k.$$

The exact marked identity says $\mathbb{E}_w \mathbb{E}_B A_B \mathcal{D}_{d+1} P = 0$ for every polynomial P of degree $d \leq N - 1$.

Lemma 1 (Global integrated spectral derivative bound). *Fix $L < \infty$. There is an absolute C (depending on the constants in (1)) such that, for every $j \geq 0$ and $0 \leq k \leq N$,*

$$\begin{aligned} (\ell + 1) \mathbb{E}_w \mathbb{E}_B A_B |a_k| & \left[(1 + |z_i - v|) \frac{|L_j^{(k)}(v)|}{k!} + \mathbf{1}_{k \geq 1} \frac{|L_j^{(k-1)}(v)|}{(k-1)!} \right] \\ & \leq C(k+1)^3 \left(C \sqrt{\frac{j+1}{m}} \right)^k. \end{aligned} \quad (2)$$

Derivatives above degree j are zero.

Proof. Write $\sigma^2 = \sum_{j \in R} (z_j - v)^2$. By (1),

$$\sigma^2 \leq \sum_{j \in R} z_j^2 \leq C(s+1)Nv.$$

The global complex-variable coefficient estimate for centered variables gives

$$|a_k| \leq \left(\frac{ek\sigma^2}{N^2} \right)^{k/2} \leq \left(\frac{Ck(s+1)v}{N} \right)^{k/2}. \quad (3)$$

For $v > 0$ the reviewed Gaussian-displacement estimate is

$$|L_j^{(h)}(v)| \leq e^{v/2}(j/v)^{h/2} \quad (0 \leq h \leq j). \quad (4)$$

This is also proved in `size_bounds/gaussian_laguerre_variance_lemmas.tex`. Using $j+1$ in place of j includes all boundary cases. If $v = 0$, all z_j in R vanish, hence $a_k = 0$ for $k \geq 1$; the estimates obtained by multiplying (3) and (4) extend to that case by continuity.

For each subset $v \leq \ell s/N$. Maclaurin's inequality gives $\mathbb{E}_B A_B \leq (1 - s/M)^\ell \leq e^{-\ell s/M}$. Consequently

$$\mathbb{E}_B A_B e^{v/2} \leq e^{-cs}$$

for an absolute $c > 0$ and sufficiently large m , since $\ell/M - \ell/(2N)$ stays positive. The Laplace bound in (1) implies, for every real $h \geq 0$,

$$(\ell+1)\mathbb{E}_w e^{-cs}(s+1)^h \leq C^{h+1}\Gamma(h+1). \quad (5)$$

For example, reserve half the exponential, bound the other half times $(s+1)^h$ by $(C(h+1))^h$, and use Stirling's inequality.

For the first derivative term in (2), (3)–(5), together with $1 + |z_i - v| \leq C(s+1)$, give the factor

$$\left(C\sqrt{\frac{j+1}{m}} \right)^k \frac{k^{k/2}\Gamma(k/2+2)}{k!} \leq C(k+1)^2 \left(C\sqrt{\frac{j+1}{m}} \right)^k.$$

For the second term, one uncanceled factor $v^{1/2}$ remains; use $v \leq Cs$ and (5). Its bound is at most

$$C(k+1)^3 m^{-1/2} \left(C\sqrt{\frac{j+1}{m}} \right)^{k-1},$$

which is absorbed into (2), since $j+1 \geq 1$. The case $k = 0$ follows directly from (4)–(5), and $k = 1$ is zero. $\square$

Theorem 1 (Arbitrarily small global residual). *Fix $0 < \delta < L < \infty$ and an exponent $A > 0$. There are constants $\kappa > 0$, $B > 0$, and $C_A > 0$ such that, for all sufficiently large m , put*

$$d = \lfloor \kappa m \rfloor, \quad J = \lceil B \log m \rceil, \quad \tau = C_A \log m/m.$$

Then $J \leq d < N$, and uniformly for $a \in [\delta, L]$,

$$|(\ell+1)\mathbb{E}_w \mathbb{E}_B A_B \mathcal{D}_J K_\tau(\cdot, a)| \leq m^{-A}. \quad (3)$$

Here K_τ is the full positive Laguerre Poisson kernel. The conclusion is global over all rows and all subsets, with no compact restriction on v and no exclusion of zero entries.

Proof. Let $r = 1 - \tau$ and

$$P_d(x) = \sum_{j=0}^d r^j L_j(a) L_j(x).$$

The Hille–Hardy identity says $K_\tau(x, a) = \sum_{j \geq 0} r^j L_j(a) L_j(x)$. The Gaussian estimate with $h = 0$ gives $|L_j(a)| \leq e^{L/2}$. The marked identity applies exactly to P_d .

Choose κ first, sufficiently small that the geometric ratio in (2) for $j \leq d$ is at most some $\rho < 1$. The contribution of differential orders $k > J$ in $\mathcal{D}_{d+1}P_d$ is then bounded by

$$\frac{C}{\tau} \sum_{k>J} (k+1)^3 \rho^k.$$

The term $k = d+1$ is allowed because $d+1 \leq N$; derivatives of P_d above its degree simply vanish. Taking B sufficiently large makes this bound at most $\frac{1}{2}m^{-A}$, even after allowing $\tau^{-1} \leq m$.

It remains to replace P_d by K_τ only at orders $k \leq J$. For each such order, Lemma 1 and $r^j \leq e^{-\tau j}$ bound the integrated absolute spectral tail by

$$C(k+1)^3 \sum_{j>d} e^{-\tau j} \left(C \sqrt{\frac{j+1}{m}} \right)^k.$$

Splitting $e^{-\tau j}$ into two factors and summing by comparison with the gamma integral gives, for $0 < \tau < 1/2$,

$$\sum_{j>d} e^{-\tau j} (j+1)^{k/2} \leq e^{-\tau d/2} C^{k+1} \Gamma(k/2 + 1) \tau^{-k/2-1}.$$

Consequently the preceding bound is at most

$$\frac{C e^{-\tau d/2}}{\tau} (k+1)^4 \left(C \sqrt{\frac{k+1}{m\tau}} \right)^k. \quad (6)$$

Choose C_A sufficiently large compared with B that the last ratio is at most $1/2$ for all $k \leq J$ and large m . The sum of (6) is therefore at most $C e^{-\tau d/2}/\tau$. Increasing C_A further so that $C_A \kappa$ exceeds a fixed multiple of $A + 2$ makes this at most $\frac{1}{2}m^{-A}$. All spectral sums of the finitely many derivatives used here are absolutely integrable by the same estimates, justifying their interchange with the two expectations. Combining the high-order truncation, the exact identity for P_d , and the low-order spectral replacement proves (3). $\square$

Theorem 2 (Unmarked companion). *With the choices in Theorem 1, define*

$$\mathcal{D}_J^0 F = \sum_{k=0}^J \frac{F^{(k)}(v)}{k!} a_k.$$

Let β_ℓ be the probability measure on $[0, \ell]$ with density

$$b_\ell(x) = \frac{\ell+1}{\ell} (1 - x/\ell)^\ell \mathbf{1}_{0 \leq x \leq \ell}.$$

Then uniformly for $a \in [\delta, L]$,

$$\left| (\ell+1) \mathbb{E}_w \mathbb{E}_B A_B \mathcal{D}_J^0 K_\tau(\cdot, a) - \int K_\tau(x, a) d\beta_\ell(x) \right| \leq m^{-A}.$$

Changing the constants to obtain the displayed coefficient one is understood. The densities b_ℓ are bounded above and below by positive constants on every fixed compact interval for large m .

Proof. For any polynomial P of degree at most N , the exact unmarked success/failure identity gives

$$(\ell+1) \mathbb{E}_w \mathbb{E}_B A_B \text{Pol}_N(P)(z_R) = (\ell+1) \int_0^1 t^\ell P(\ell(1-t)) dt = \int P d\beta_\ell.$$

Apply it to P_d . Lemma 1 bounds its differential-order truncation and the low-order spectral replacement on the row side, exactly as in Theorem 1. On the reference side $b_\ell(x) \leq 2e^{-x}$ for large m , so $\int e^{x/2} d\beta_\ell \leq 4$. Consequently the reference spectral tail is at most $Ce^{-\tau d}/\tau$. Increasing the constants in the previous proof, or running that proof with exponent $A+1$, completes the estimate. The last assertion follows directly from the displayed density and $(1-x/\ell)^\ell \rightarrow e^{-x}$ uniformly on compact sets. $\square$

Lemma 2 (Uniform complex bound). *For $a \in [\delta, L]$, $0 < \tau < 1/2$, $v \geq 0$, and $|\zeta| \leq \tau$,*

$$|K_\tau(v + \zeta, a)| \leq C_L/\tau.$$

Thus $|\partial_v^k K_\tau(v, a)|/k! \leq C_L \tau^{-k-1}$.

Proof. The entire power series gives $|I_0(z)| \leq e^{|z|}$. Write $r = 1 - \tau$ and $y = v + \tau$. The exponent in this bound for $K_\tau(v + \zeta, a)$ is at most

$$\frac{-ry - ra + 2r\tau + 2\sqrt{ray}}{\tau} \leq \frac{a - ra + 2r\tau}{\tau} \leq L + 2,$$

where $-ry + 2\sqrt{ray} \leq a$ follows by completing a square. Cauchy's formula proves the derivative assertion. $\square$

Theorem 3 (Off-annulus differential tails). *For every fixed $0 < \delta < L$ there are constants $0 < s_- < s_+ < \infty$ such that, with the choices above, the contribution to either*

$$(\ell + 1)\mathbb{E}_w \mathbb{E}_{BA_B} |\mathcal{D}_J K_\tau|, \quad (\ell + 1)\mathbb{E}_w \mathbb{E}_{BA_B} |\mathcal{D}_J^0 K_\tau|$$

from rows $s \notin [s_-, s_+]$ is at most m^{-A} for sufficiently large m . The same statement holds if absolute values are placed on every summand of either differential functional separately.

Proof. For $v \leq \delta/16$ and $|\zeta| \leq \delta/16$, the elementary bound $|I_0(z)| \leq e^{|z|}$ gives

$$|K_\tau(v + \zeta, a)| \leq \tau^{-1} e^{-c\delta/\tau}.$$

Indeed its exponent is at most $[r\delta/16 - ra + 2\sqrt{ra\delta/8}]/\tau$, which is bounded above by $-c\delta/\tau$ for $a \geq \delta$ and sufficiently small τ . For $v \geq 16L$ and $|\zeta| \leq v/16$, the same argument gives

$$|K_\tau(v + \zeta, a)| \leq \tau^{-1} e^{-cv/\tau}.$$

Cauchy's formula gives the corresponding derivative bounds, with radii $\delta/16$ and $v/16$ respectively.

Take s_- sufficiently small that $v \leq \ell s/N \leq \delta/16$ whenever $s \leq s_-$. By (1) the row range and marked value are bounded on those rows. The coefficient estimate $|a_k| \leq (CH\sqrt{k/N})^k$ therefore makes the sum of the Cauchy bounds through $J = B \log m$ at most $C\tau^{-1}e^{-c\delta/\tau}$. The marked factors are bounded as well. Multiplication by $\ell + 1 \leq m$ leaves a quantity smaller than any prescribed inverse power of m .

For $s \geq s_+$, choose s_+ large enough that $x = (\ell/M)s$ satisfies $x/2 \geq 16L$. On subsets with $v \geq x/2$, the row range and marked value are $O(s+1) = O(v)$ by (1). The large- v Cauchy radius thus cancels the range in the coefficient estimate. Summing through J gives at most a polynomial factor in m times $e^{-cs/\tau}$, again negligible even after integration against the probability w .

For completeness the exceptional subsets $v < x/2$ have uniform probability at most e^{-cm} . To see this without assumptions on the success weights, sample the N -subset R uniformly. Maclaurin's inequality applied to the positive numbers $e^{t(z_j - x)/N}$, followed by the elementary bounded-variable exponential estimate, gives

$$\mathbb{E}_R e^{t(v-x)} \leq e^{t^2 H^2 / (8N)}.$$

Here H is the unmarked population range. Hence $\Pr(v < x/2) \leq \exp[-Nx^2/(2H^2)] \leq e^{-cm}$, because $H \leq C(s+1)$ and $s \geq s_+ > 0$. On these subsets use Lemma 2, not a full high-order polarization bound. Since $s \leq M$, all z_j are $O(m)$, and for $k \leq J = B \log m$ the product of the crude coefficient bound and τ^{-k-1} is at most $\exp(C_B(\log m)^2)$. The marked prefactors and $\ell+1$ are polynomial in m , while $A_B \leq 1$. Thus the exceptional-subset contribution is at most $e^{-cm} \exp(C_B(\log m)^2)$, which is negligible. This proves all assertions. $\square$

Order of choices and scope. The constants are chosen in the order κ, B , then C_A . They do not depend on m . The theorem supplies a global analytic residual for the marked identity. Turning it into a sharp profile bound still requires positivity/localization of the full kernel, control of the success-weighted subset mean, and a monotonicity argument. None of those subsequent conclusions is asserted here.